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# Level-2 Data Flow Diagram and Database Schema

## E-Commerce Web Application

### 1. Basis and Scope

This document extends the existing E-Commerce Web Application requirements into a Level-2 Data Flow Diagram (DFD) and a logical relational database schema. The scope follows the SRS modules: user authentication, product browsing, shopping cart management, checkout/order creation, order history, and admin product/inventory management.

The Level-2 DFD decomposes major Level-1 processes into lower-level sub-processes and shows the external entities, data stores, and named data flows required for implementation in a Django MVT application.

### 2. DFD Notation Used

| DFD Element     | Notation in This Report | Meaning                                                                   |
|-----------------|-------------------------|---------------------------------------------------------------------------|
| External Entity | Yellow rectangle        | Actor/system outside the application boundary that sends or receives data |
| Process         | Blue rounded rectangle  | Business activity that transforms input data into output data             |
| Data Store      | Green cylinder          | Persistent repository or database-backed table                            |
| Data Flow       | Directed arrow          | Named movement of data between entities, processes, and stores            |

### 3. Level-2 DFD

The Level-2 DFD is split into two readable sub-diagrams. Figure 1 covers authentication, product browsing, and cart management. Figure 2 covers checkout, order history, and admin product/inventory management.

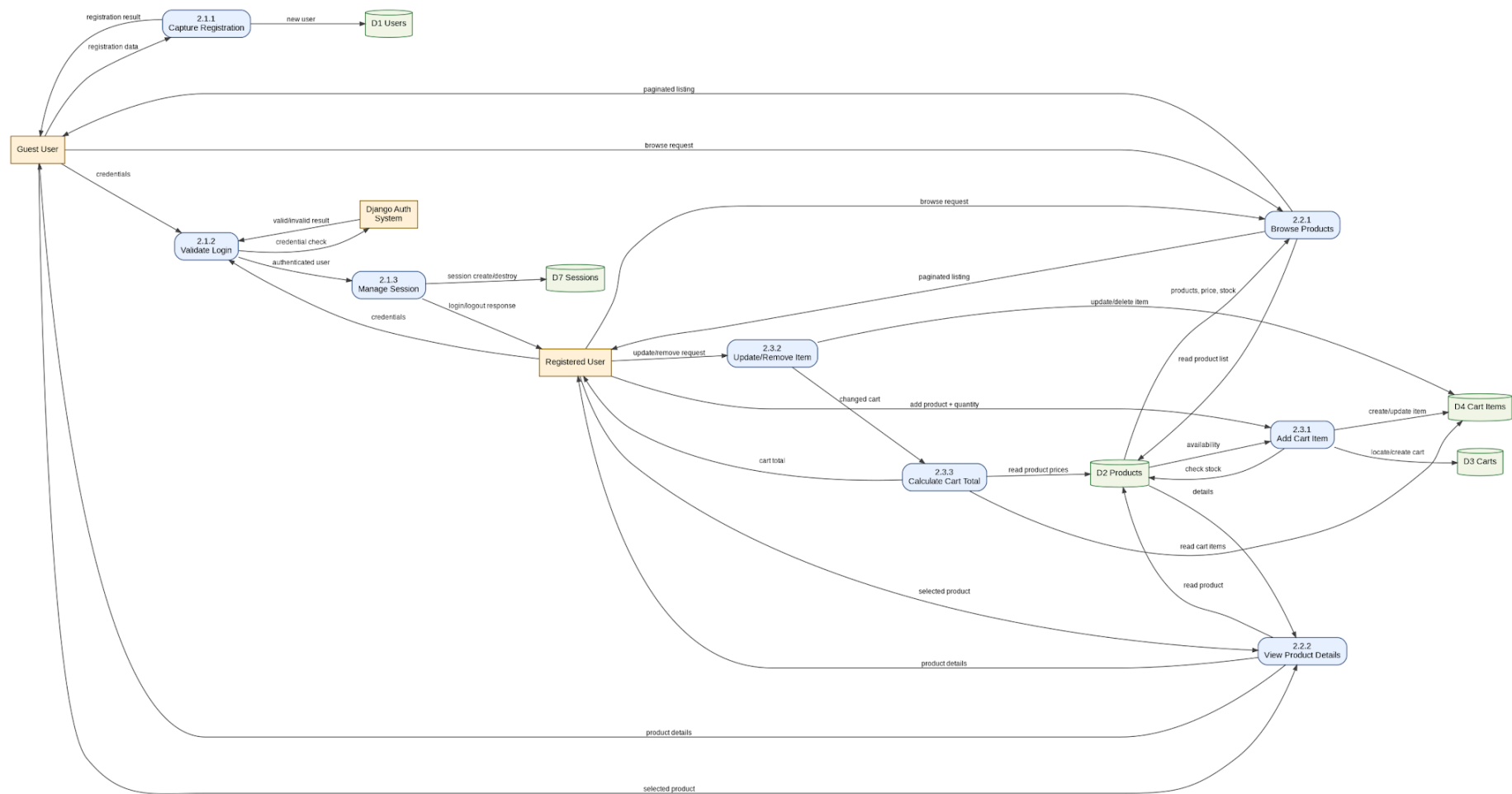


Figure 1: Level-2 DFD - Authentication, Product Browsing, and Cart Management

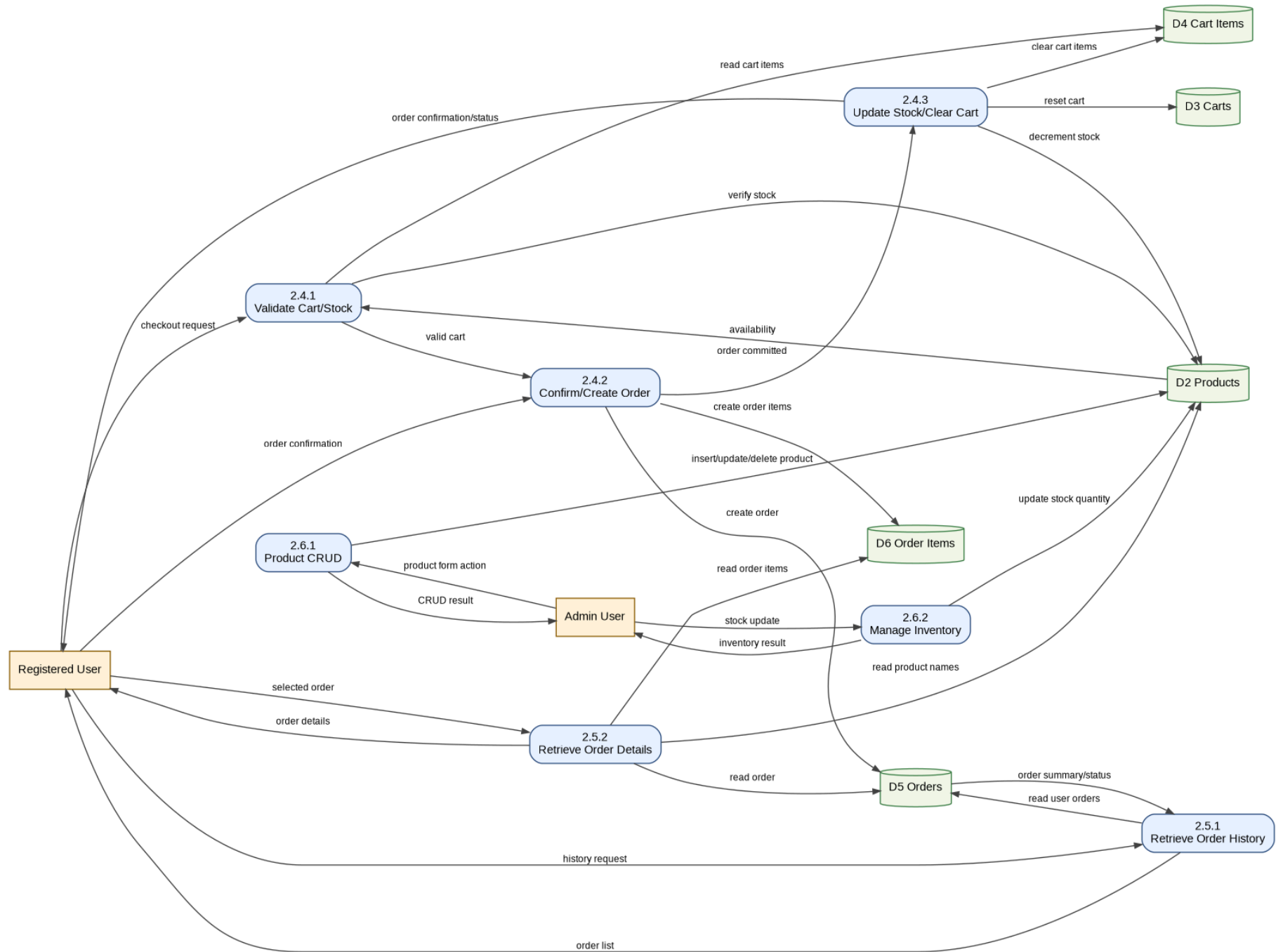


Figure 2: Level-2 DFD - Checkout, Orders, and Admin Management

| Level-2 Process               | Description                                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------------------|
| 2.1.1 Capture Registration    | Receives username, email, password, and confirmation; creates a user record after validation. |
| 2.1.2 Validate Login          | Checks credentials through Django authentication and returns valid/invalid login status.      |
| 2.1.3 Manage Session          | Creates, maintains, or destroys the authenticated user session.                               |
| 2.2.1 Browse Products         | Retrieves product list, price, stock, and pagination-ready product summary data.              |
| 2.2.2 View Product Details    | Retrieves selected product description, image, price, and availability.                       |
| 2.3.1 Add Cart Item           | Checks stock and creates or updates a CartItem for the user cart.                             |
| 2.3.2 Update/Remove Item      | Updates quantity or removes an item from the cart.                                            |
| 2.3.3 Calculate Cart Total    | Reads cart items and product prices to calculate the dynamic total.                           |
| 2.4.1 Validate Cart/Stock     | Verifies that the cart is non-empty and ordered quantities are available.                     |
| 2.4.2 Confirm/Create Order    | Creates Order and OrderItem records after user confirmation.                                  |
| 2.4.3 Update Stock/Clear Cart | Reduces product stock and clears cart items after successful order creation.                  |
| 2.5.1 Retrieve Order History  | Retrieves all past orders for the logged-in user.                                             |
| 2.5.2 Retrieve Order Details  | Retrieves selected order, order items, product names, total, date, and status.                |
| 2.6.1 Product CRUD            | Allows admin users to create, update, or delete products.                                     |
| 2.6.2 Manage Inventory        | Allows admin users to update stock quantities and product availability.                       |

## 4. Data Stores

| Store ID | Data Store  | Main Data Held                                              | Used By                                         |
|----------|-------------|-------------------------------------------------------------|-------------------------------------------------|
| D1       | Users       | User ID, username, email, hashed password, staff/admin flag | Authentication, order ownership, cart ownership |
| D2       | Products    | Product name, description, image, price, stock quantity     | Browsing, cart, checkout, admin inventory       |
| D3       | Carts       | Cart ID and owner user                                      | Cart management and checkout                    |
| D4       | Cart Items  | Cart-product lines and quantities                           | Cart total, checkout validation, cart updates   |
| D5       | Orders      | Order header: user, total, date, status                     | Checkout and order history                      |
| D6       | Order Items | Purchased items, quantities, unit prices                    | Order detail and receipt/history views          |
| D7       | Sessions    | Authenticated session data                                  | Login/logout and protected operations           |

## 5. Key Data Flow Summary

| Flow Area        | Input Data                                  | Output Data / Store Update                                              |
|------------------|---------------------------------------------|-------------------------------------------------------------------------|
| Registration     | Username, email, password, confirm password | Validated User record in D1; registration result returned to Guest User |
| Login/Logout     | Credentials or logout request               | Auth result and session create/destroy in D7                            |
| Product Browsing | Browse request or selected product ID       | Product list/detail data read from D2                                   |

|                          |                                            |                                                                               |
|--------------------------|--------------------------------------------|-------------------------------------------------------------------------------|
| Cart Management          | Product ID, quantity, update/remove action | Cart and CartItem updates in D3/D4; calculated total returned to user         |
| Checkout                 | Cart contents and order confirmation       | Order/OrderItem records in D5/D6; stock decremented in D2; cart cleared in D4 |
| Order History            | Authenticated user and selected order ID   | Order summary/detail with items and status                                    |
| Admin Product Management | Product form data or stock update          | Product create/edit/delete and inventory update in D2                         |

## 6. Database Schema

The schema is a normalized relational model suitable for Django ORM implementation. It follows the SRS logical database requirements while adding practical constraints such as unique usernames/emails, non-negative stock, positive quantities, and controlled order statuses.

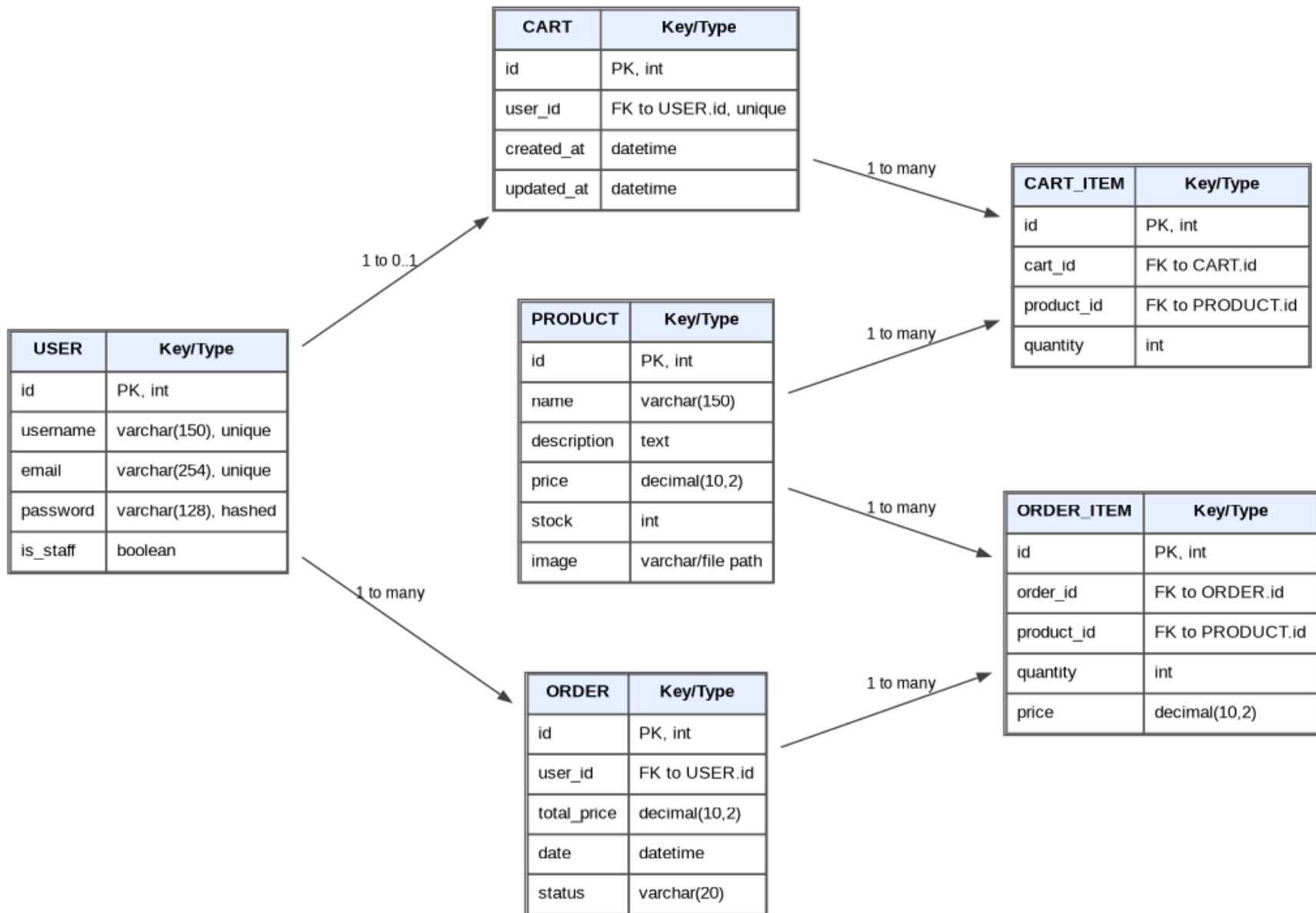


Figure 3: Logical Database Schema / ERD for the E-Commerce Web Application

| Table     | Fields                                                                       | Purpose                                                                                                    |
|-----------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| User      | id PK; username; email; password; is_staff                                   | Stores registered user/admin credentials and authorization flag. Password is stored hashed, not plaintext. |
| Product   | id PK; name; description; price; stock; image                                | Stores sellable product information and inventory quantity.                                                |
| Cart      | id PK; user_id FK to User.id; created_at; updated_at                         | Represents the active shopping cart for a registered user.                                                 |
| CartItem  | id PK; cart_id FK to Cart.id; product_id FK to Product.id; quantity          | Stores selected products and quantities before checkout.                                                   |
| Order     | id PK; user_id FK to User.id; total_price; date; status                      | Stores a confirmed purchase transaction and current order status.                                          |
| OrderItem | id PK; order_id FK to Order.id; product_id FK to Product.id; quantity; price | Stores immutable line-item details for an order at purchase time.                                          |

## 7. Relationships and Cardinality

| Relationship         | Cardinality | Explanation                                                                     |
|----------------------|-------------|---------------------------------------------------------------------------------|
| User to Cart         | 1 to 0..1   | A registered user may have one active cart; a cart belongs to exactly one user. |
| Cart to CartItem     | 1 to many   | A cart can contain multiple product line items.                                 |
| Product to CartItem  | 1 to many   | A product can appear in many users carts.                                       |
| User to Order        | 1 to many   | A user can place many orders; each order belongs to one user.                   |
| Order to OrderItem   | 1 to many   | An order contains one or more purchased product lines.                          |
| Product to OrderItem | 1 to many   | A product can appear in many historical order line items.                       |

## 8. Constraints and Validation Rules

| Rule                                | Applied To                               | Reason                                                                      |
|-------------------------------------|------------------------------------------|-----------------------------------------------------------------------------|
| username must be unique             | User.username                            | Supports unambiguous authentication and registration validation.            |
| email should be unique              | User.email                               | Prevents duplicate customer identity records.                               |
| password must be hashed             | User.password                            | Protects credentials and satisfies security requirements.                   |
| price >= 0                          | Product.price and OrderItem.price        | Prevents invalid product/order pricing.                                     |
| stock >= 0                          | Product.stock                            | Prevents negative inventory after checkout.                                 |
| quantity > 0                        | CartItem.quantity and OrderItem.quantity | Prevents empty or invalid cart/order lines.                                 |
| status in Pending/Shipped/Delivered | Order.status                             | Restricts order lifecycle values to the required statuses.                  |
| admin-only product CRUD             | Product management processes             | Ensures only privileged users can create, edit, delete, or manage products. |

## 9. Suggested Django Model Mapping

```
class Product(models.Model):
```

```
    name = models.CharField(max_length=150)
```

```
    description = models.TextField()
```

```
    price = models.DecimalField(max_digits=10, decimal_places=2)
```

```
    stock = models.PositiveIntegerField(default=0)
```

```
    image = models.ImageField(upload_to="products/", blank=True, null=True)
```

```
class Cart(models.Model):
```

```
    user = models.OneToOneField(User, on_delete=models.CASCADE)
```

```
    created_at = models.DateTimeField(auto_now_add=True)
```

```
    updated_at = models.DateTimeField(auto_now=True)
```

```
class CartItem(models.Model):
```

```
    cart = models.ForeignKey(Cart, on_delete=models.CASCADE, related_name="items")
```

```
    product = models.ForeignKey(Product, on_delete=models.CASCADE)
```

```
    quantity = models.PositiveIntegerField(default=1)
```

```
class Order(models.Model):
```

```
    STATUS_CHOICES = [("Pending", "Pending"), ("Shipped", "Shipped"), ("Delivered", "Delivered")]
```

```
    user = models.ForeignKey(User, on_delete=models.CASCADE)
```

```
    total_price = models.DecimalField(max_digits=10, decimal_places=2)
```

```
    date = models.DateTimeField(auto_now_add=True)
```

```
    status = models.CharField(max_length=20, choices=STATUS_CHOICES, default="Pending")
```

```
class OrderItem(models.Model):
```

```
order = models.ForeignKey(Order, on_delete=models.CASCADE, related_name="items")

product = models.ForeignKey(Product, on_delete=models.PROTECT)

quantity = models.PositiveIntegerField()

price = models.DecimalField(max_digits=10, decimal_places=2)
```

## 10. Traceability to Requirements

| Requirement Area                | FR References | DFD Processes       | Database Tables                     |
|---------------------------------|---------------|---------------------|-------------------------------------|
| Registration and authentication | FR1-FR8       | 2.1.1, 2.1.2, 2.1.3 | User, Sessions                      |
| Admin product management        | FR9-FR12      | 2.6.1, 2.6.2        | Product                             |
| Product browsing                | FR13-FR16     | 2.2.1, 2.2.2        | Product                             |
| Shopping cart                   | FR17-FR21     | 2.3.1, 2.3.2, 2.3.3 | Cart, CartItem, Product             |
| Checkout and orders             | FR22-FR25     | 2.4.1, 2.4.2, 2.4.3 | Order, OrderItem, Product, CartItem |
| Order history                   | FR26-FR28     | 2.5.1, 2.5.2        | Order, OrderItem, Product           |